Analysis of the Historic Electricity Generation Mix Data

Introduction

In 2024, the UK government released the "Clean Power 2030 Action Plan". This plan, titled "A New Era for Clean Electricity" was published by the Department for Energy Security and Net Zero (DESNZ). This plan calls for decarbonisation of UK's power generation system by the end of the decade. The plan sets out targets and timelines to achieve those targets by 2030, in order to have a more carbon neutral grid.

In this report, I have used Neso's Historic Generation Mix dataset to explore the evolution of the fuel composition for electricity generation over time. This data is very detailed with half-hour timestamps and has 34 columns detailing the instantaneous power being generated from various sources as well as the percentage contribution of each source in the total generation mix.

I downloaded data from https://www.neso.energy/data-portal/historic-generation-mix.

The Neso website gives the following description:

"The carbon intensity of electricity measures how much carbon dioxide emissions are produced per kilowatt hour.
This dataset contains data from 1 January 2009. It has seasonal decomposition applied to correct missing or irregular data points. The data is subject to change due to a data cleansing process taking place to provide the most accurate figures.

The Historic Generation Mix dataset (full title: Historic GB Generation Mix) from the National Energy System Operator (NESO) represents historical electricity generation data for Great Britain (GB). It provides details on the mix of fuel types/sources used to generate electricity, along with associated carbon intensity (grams of CO_2 emitted per kilowatt-hour of electricity)."

The datapoints are revised on a regular basis.

Explanation of the Columns of the Data

This data has 34 columns and the key aspects it covers are :

- **Generation by fuel type:** This includes how much electricity came from different sources (e.g., renewables like wind/solar/biomass, gas (such as CCGT), coal, nuclear, imports, etc.), reported in megawatts (MW) (instantaneous power) or percentages (%) of the total mix. Most of the column names are self-explanatory. However, for the sake of clarity, some of the data-values which might not be obvious are explained below.
- **Fossil:** Power generated from coal and gas.
- **Renewable:** Power generated from the following sources - wind, wind_emb ,hydro , biomass and solar. The column **wind_emb** represents the power generated from smaller wind farms incorporated into the local distribution network but not the high voltage grid. These are typically smaller onshore wind installations (farms or single turbines). Their output isn't metered at the transmission level by Neso, so it's estimated using models (based on weather data, installed capacity, historical patterns, etc.). Embedded wind production reduces demand on the transmission grid by offseting local demand so it’s added to the generation mix to give an idea of the contribution of these smaller wind farms to the total power generation in UK.
- **Zero carbon_perc:** The percentage of total power generated from nuclear, wind, wind_emb, hydro and solar sources
- **Low carbon:** zero carbon souces and power generated from biomass fuel
- **Imports:** This covers the sum of power that is imported from other countries, notably from Netherlands, Ireland, Norway, Denmark etc. This is net power from all imports and can be negative if energy is being exported.
- **Biomass:** This column represents the power generated from biomass plants (burning wood pellets, chips, agricultural residue)
- **Other:** This is the power generated from smaller or miscellaneous sources not fitting into the main named fuels (gas, coal, nuclear, wind, etc.). This includes landfill gas, sewage gas, and biogas engines etc. It’s a very small percentage of the total power generated in the mix.
- **Generation:** This is the sum of power generated from gas, coal, nuclear, wind, hydro and imports
- **Carbon intensity:** Measures the CO_2 emissions associated with the generation mix – the grams of CO_2 emitted from 1 kWh of energy generated (gCO_2/kWh).

- The data starts from 1 January 2009 and extends to the present (continuously updated).
- It is half-hourly resolution (common for GB electricity data, though the portal describes updates as hourly in some metadata contexts; actual data points are typically every 30 minutes).
- The data undergoes ongoing cleansing and is subject to potential revisions for accuracy.
- Geographical scope: Great Britain (England, Scotland, Wales; excludes Northern Ireland, which has a separate system).

Objectives

- To analyse the evolution of electricity generation in UK from 2009 onwards focussing on changes in total output and contribution of different sources over time.
- To verify how the carbon emissions due to electricity generation have changed over time?(keeping in mind the UK government's aims for fulfilling 95% its electricity needs from clean sources (renewables, nuclear, and future technologies like gas with carbon capture))
- What trend can be observed in UK's increase in adoption of renewable sources (hydro, wind, biomass) for power generation since renewable sources is one of the foundations of achieving net-zero emissions? (despite the fact that renewable sources might not be inherently carbon-neutral).
- Is UK on track to reduce its carbon emissions from power generation to 50g/kWh as stated in the 'Clean Power 2030 Action Plan'? (The government's plan is to decrease emissions from 171g/kWh (2023 levels) to well below 50g/kWh by 2030)
- To what extent is UK progressing towards the government's plan to fulfill 95% of its total energy needs from low carbon sources (renewables, nuclear)?

The Process of Analysis:

After downloading the data, I loaded it in PowerBI. I transformed the datatypes from general to either whole numbers or decimal depending on column values. The date-time column was converted to date-time datatype. There were no null values or blanks in the dataset.

Creating Measures:

In order to achieve my objectives and make suitable visualisations, I calculated the following measures- first the averages of the columns: BIOMASS_perc, COAL_perc, WIND_EMB_perc, CARBON_INTENSITY, FOSSIL_perc, GAS_perc, HYDRO_perc, HYDRO_perc, IMPORTS_perc, LOW_CARBON_perc, NUCLEAR_perc, OTHER_perc, RENEWABLE_perc, SOLAR_perc, STORAGE_perc, WIND_perc, ZERO_CARBON_perc. I also calculated the measure of Renewable Energy in gigawatt-hours and used this measure to calculate the year-over-year percentage change in renewable energy production. Apart from the average of percentages of generation by fuel type columns, I also calculated the measures which are averages of the power generated by FOSSIL, wind(average of the sum of columns WIND and WIND_EMB), HYDRO, NUCLEAR, IMPORTS, BIOMASS, SOLAR, STORAGE,GENERATION, and OTHER. I also made a seperate table named 'Calendar' with all the dates and times from the original column. I modeled the data so that the table 'df_fuel_ckan' and 'Calendar' have a 1 to 1 relationship mapped on the datetime column. This was to create a heirarchy in the datetime column to produce accurate plotting.

Creating Visualisations and Understanding the Data to Achieve the Above Objectives:

After making the measures, I then plotted the visualisations – the first one was to see how the usage of fossil fuels have changed over the years in relation to the total generation mix. I initially plotted a stacked bar chart using the measures created above gas, coal, nuclear, wind, hydro and imports against years. This visualisation was not adding up to a hundred percent causing confusion.
To make sense of these graphs not adding up to a hundred percent, I made another calculate column named "Core Sum of Energy Mix" which added the column values of the percentages of power generation from gas, coal, nuclear, hydro, import, wind and wind_emb. This was to check if the percentage generation column is actually a sum of what's stated by NESO as "Sum of gas, coal, nuclear, wind, hydro and imports." It turns out that values of this calculated column and generation percentage column has a differences of ~ 0.1 to 0.2 % in the earlier years from 2009 till about 2014 and then the discrepancy between the values of the generation percentage and my calculated column increased by 4 – 5 % and even more as I went further down the years. I then added more column values to my calculated column - the percentage columns namely: OTHER_perc, BIOMASS_perc, STORAGE_perc and 'SOLAR_perc'. I then calculated a measure of the maximum absolute difference in the column 'GENERATION_perc' and my calculated column 'Sum of Individual Energy% Mix' which came out to be much closer to the 'GENERATION_perc' column. I am deducing that although NESO website states differently,

the ‘GENERATION_perc’ column is a sum of power generated from all sources correctly showing that 100% correlates to the power demands of UK. The average difference in the calculated column and ‘GENERATION_perc’ column is 0.3 % and the maximum is 1.7%. These are the calculated measures. I am deducing that this difference is due to rounding some columns when added together for the ‘GENERATION_perc’ column. I then added these measures to my stacked column chart to get an accurate snapshot of how the percentage contribution of each fuel type in electricity generation is changing over the years. From the chart in Figure 1, it can be seen clearly that since 2013, electric power

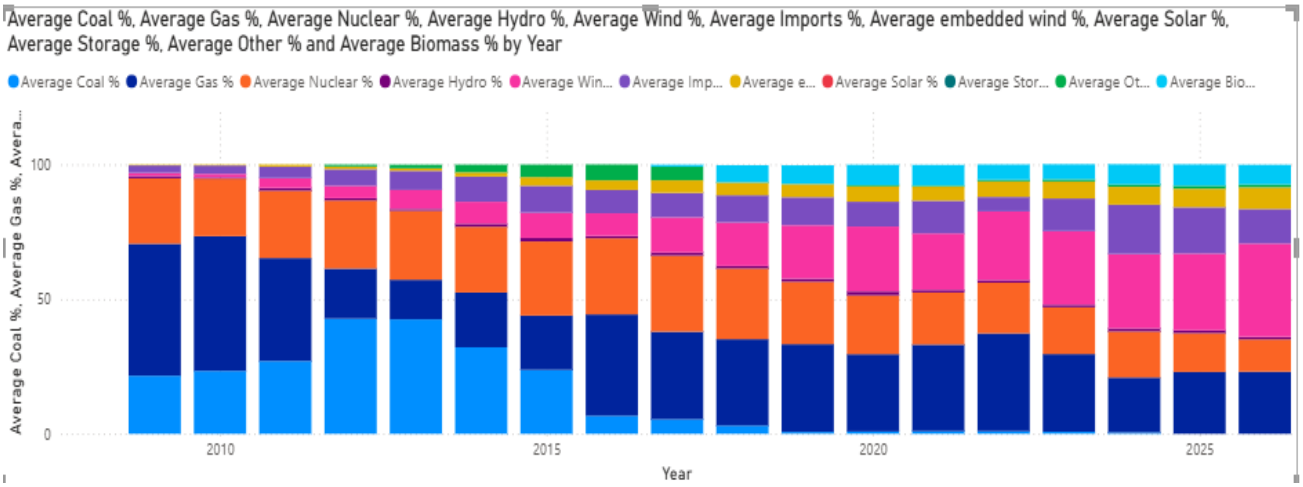


Figure 1

generation using coal have levelled off. Electricity generation using gas has been decreasing since 2022. This chart gives a true reflection of the power generation contributed by the different sectors involved as it gives the percentage of power generation by different sectors while encompassing the varying total power demand of the country.

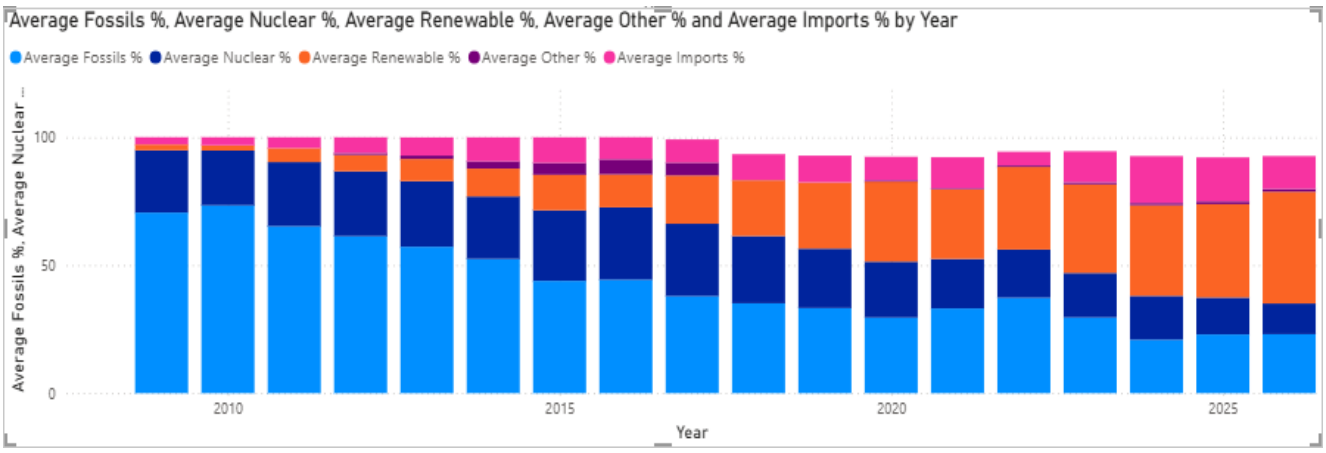


Figure 2

In the above Figure 2, I have plotted the average percentage power generated using fossil fuels, nuclear (low carbon emission), renewable (low carbon emission), average of others and imports. This graph shows the change in power generation using renewable energy which is a sum of hydro, wind, embedded wind, biomass and solar. On the whole, it produces low carbon emission, but it is a cornerstone of UK's policy of moving towards reducing carbon emissions (more than 100% of energy to be generated from clean sources, which is renewable, nuclear , gas with carbon capture) by 2030. As can be seen from Figure 2 that fossil fuel power generation is slowly levelling off with renewables and nuclear generation (which is low carbon emission, but is not renewable) increasing. From Figure 1, it can be seen that power generation from Wind is increasing - a good sign. I also added drill through feature in the Power BI report to see the monthly fluctuations.

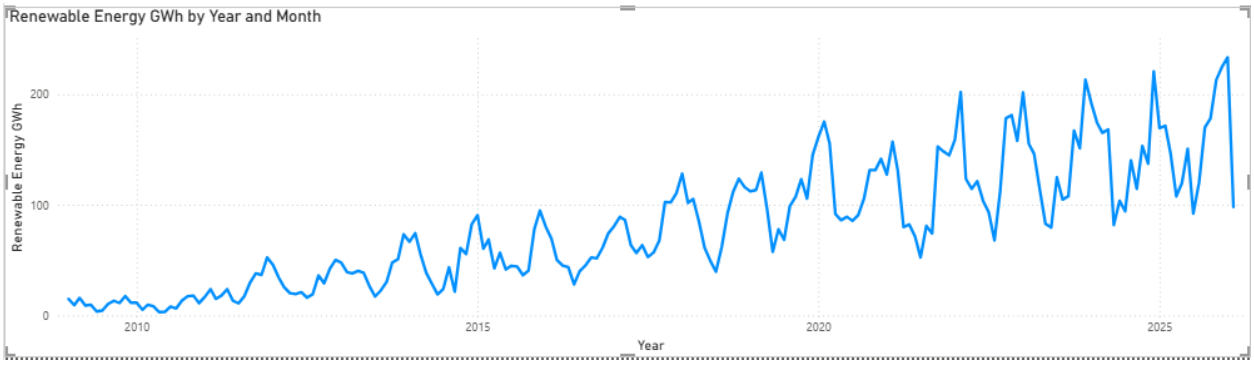


Figure 3

The above shows the upward trend in renewable energy generation plotted from the renewable energy measure I calculated earlier. Each year the Renewable energy generation peaks in December, January or February.

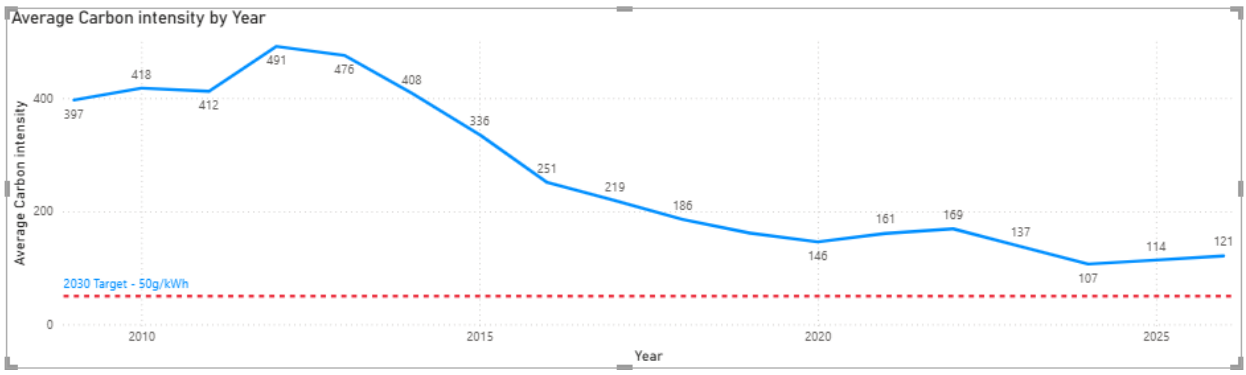


Figure 4

Figure 4 shows the change in carbon intensity (g/kWh of carbon released in the atmosphere) with time. There's a downward trend in the carbon emissions over time but as can be seen in the graph, the carbon intensity is as of 2025 is 114g/kWh and still has to

decrease quite a bit to reach the 2030 target level of 50g/kWh as can be seen from the target red-dotted line.

The graph in Figure 5 shows the change in average low carbon electricity generation and Total Power generation over time. Again, I've indicated the government's target of

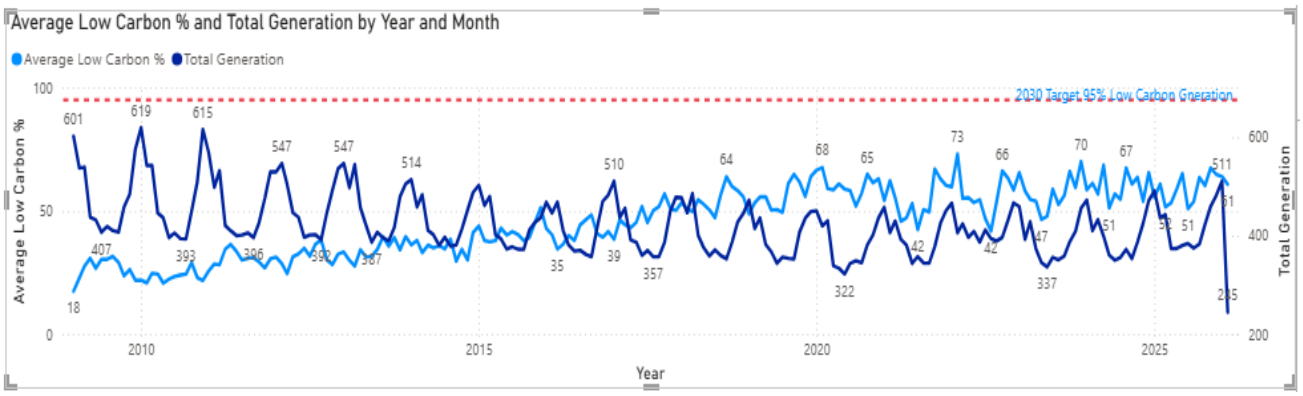


Figure 5

generating 95% of its total electric power needs from low-carbon sources with the dotted red line. Apart from monthly fluctuations, the percentage generation has not made much progress in the last 5 years.

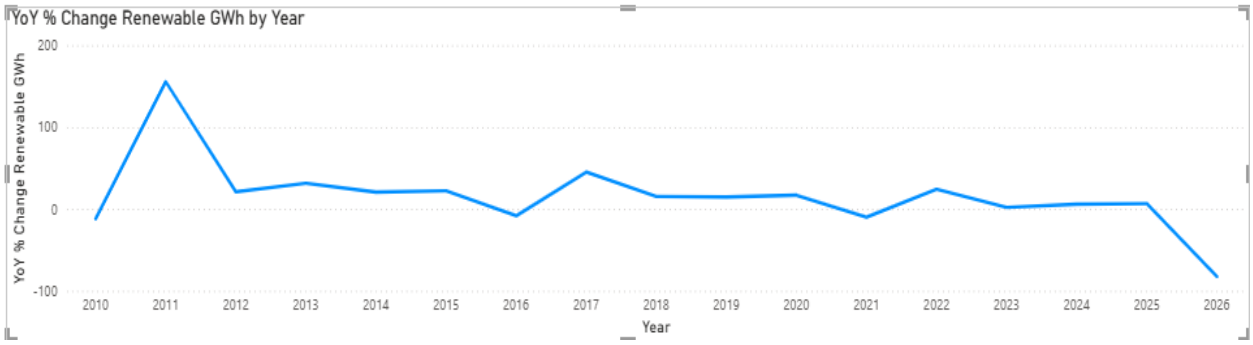


Figure 6

The figure above shows the year-over-year percentage change in renewable energy production. It's mostly positive showing the gradual upward trend.

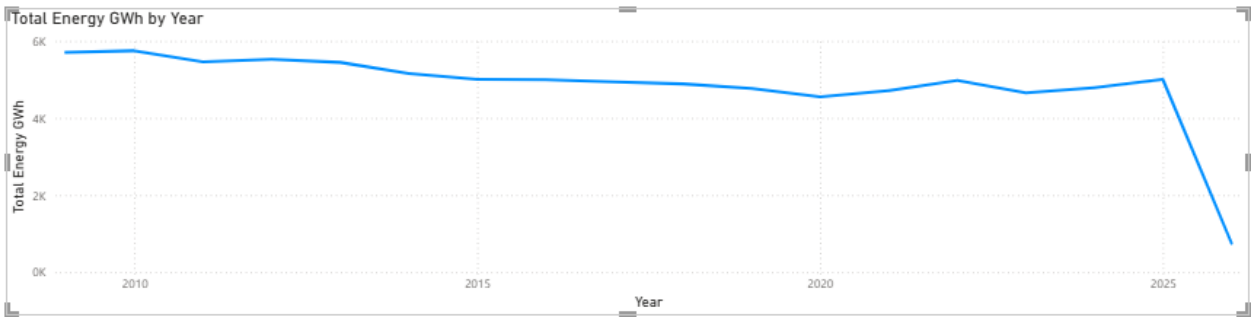


Figure 7

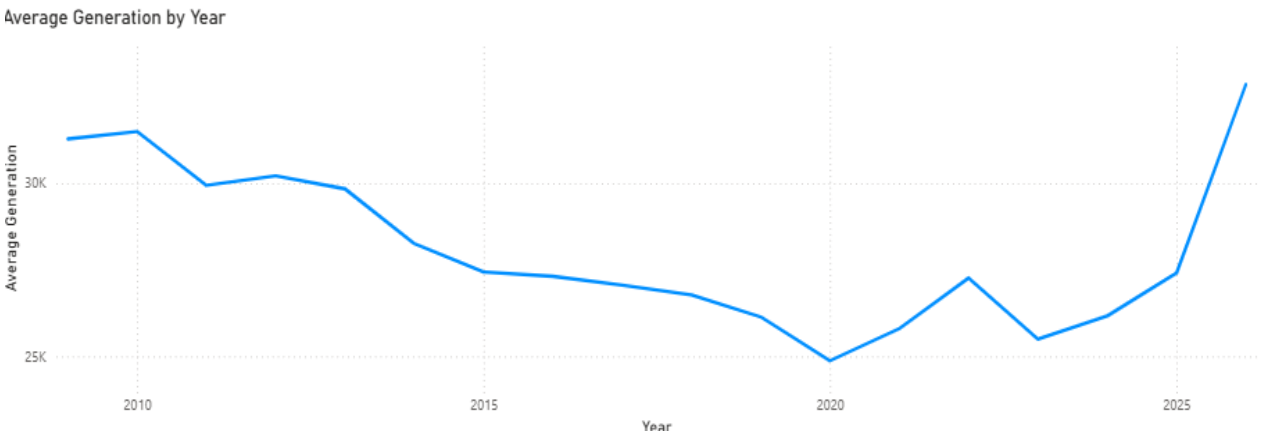
This last graph in Figure 7 shows the total energy needs of UK over the last 16 years. The electrical energy consumed in UK has decreased since 2010 due to increasing efficiency. There is a slight increase from 2020 till 2022 due to covid and work from home policies, a decrease in demand from 2022 to 2023 and again an increase from 2023 onwards.

Achievement of Objectives

I will now discuss how I've achieved the objectives I defined in the beginning of this analysis.

1.To analyse the evolution of electricity generation in UK from 2009 onwards focussing on changes in total output and contribution of different sources over time.

Since 2009, there has been an overall decline in the electrical energy demands in UK. In 2009, the average electrical energy utilised in UK was 5708.63 GWh. In 2025, it was 5002GWh. There has been a decrease in the electrical energy demands of UK but the demand in last couple years the demand is picking up again. Britian has successfully

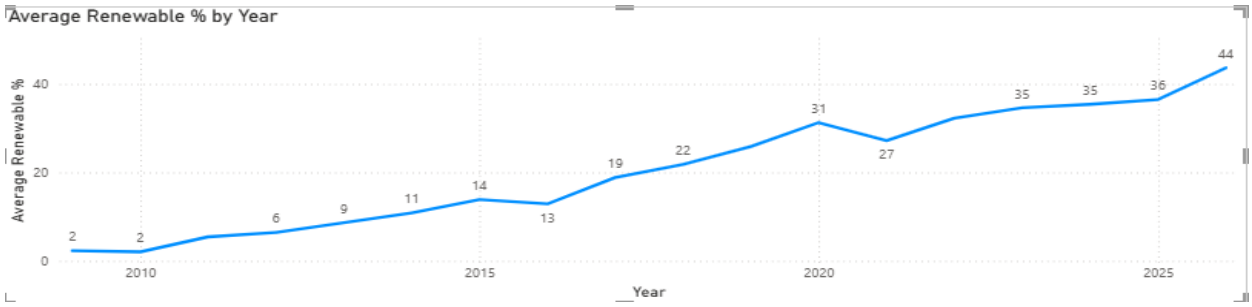


decreased the percentage of fossil fuels used in electricity generation from 73.38% in 2010 to 22.97% in 2025 – a sizeable achievement. The power generated from wind has gone up from 1.61% of the total fuel mix in 2010 to 35.35% again a solid improvement. The electricity generated from renewable sources has gone up from 2.07% in 2010 to 36.50% in 2025. The details of the contribution of each fuel source in the total mix can be seen from the Power BI file attached.

2.To verify how the carbon emissions due to electricity generation have changed over time?

The average carbon emission in 2009 was 396.65 g/kWh, peaking in 2012 to 491.48 g/kWh to an all-time low of 106.66 g/kWh in 2024 and increasing a bit to113.78 g/kWh in 2025. The details of the trend can be seen in the above Figure 4.

3.What trend can be observed in UK's increase in adoption of renewable sources (hydro, wind, biomass) for power generation since renewable sources is one of the foundations of achieving net-zero emissions?



The electricity generated from renewable sources has gone up from 2.07% in 2010 to 36.50% in 2025 as seen in the graph above. These values represent the power generated as a percentage of the total power demand in UK. The steady upward trend is clearly visible.

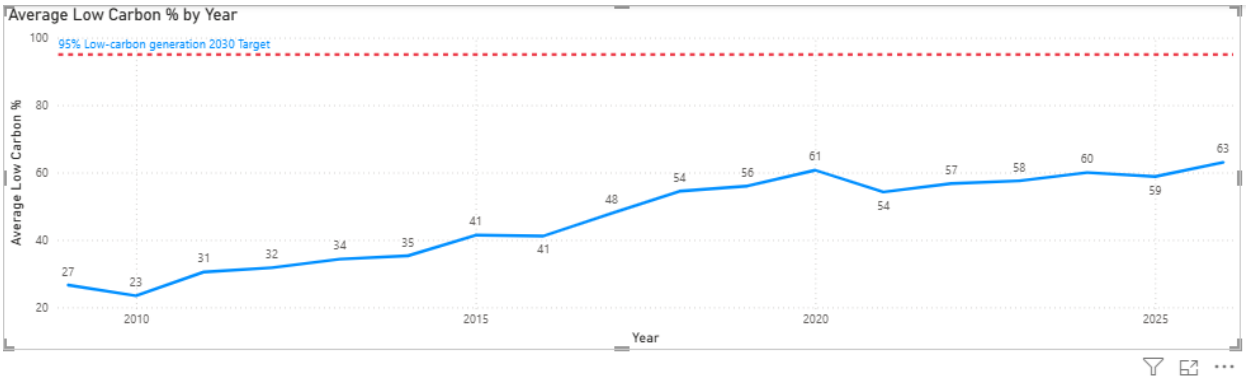
4.Is UK on track to reduce its carbon emissions from power generation to 50g/kWh as stated in the 'Clean Power 2030 Action Plan'?

The Clean Power 2030 mission defined by the government has set the target to reduce the carbon emissions to a level below 50g/kWh by 2030. As can be seen from figure 4, the carbon emissions have decreased from 397 g/kWh to 114 g/kWh in 2025. This is a 71% decrease in 16 years. There is an increase from 2024 to 2025 from 106 g/kWh to 114 g/kWh showing that this decrease is not consistent. It is still more than twice the level of the target emissions of 50 g/kWh which is to be achieved by 2030.

According to the data above, the government's target for 2030 is quite ambitious and keeping in view the past trajectory, UK as of now is **not** on track to meet the 2030 target unless some drastic measures are taken.

5.To what extent is UK progressing towards the government's plan to fulfill 95% of its total energy needs from low carbon sources (renewables, nuclear)?

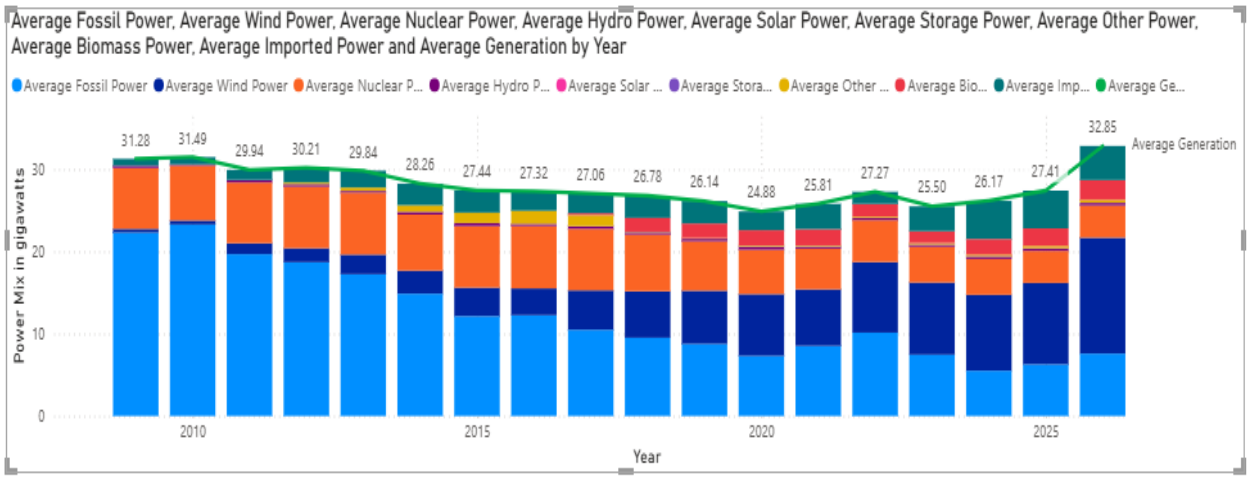
The sources that have low carbon emissions are wind, solar, nuclear, hydropower, and biomass.



Again, in the graph above, the upward trend of the penetration of electricity generation using low-carbon sources can clearly be seen. From just 27% in 2009 to 59% in 2025, there is a steady increase in the usage of low-carbon sources for power generation.

This is a sizeable increase but if you look at the past five years, the percentage of power generated from renewable sources has actually decreased from 61% to 59% in 2025.

This decrease in the rate of low-carbon sources in the past 5 years does not forecast that UK will achieve the target of producing 95% of its total electrical energy needs from low-carbon sources.



This last figure above shows the proportion of UK energy mix in proportion to the changing total generation. The previous bar chart in Figure 1 and Figure 2 showed the share of different fuel sources as percentages of the the total generation. The above chart shows the changing power demand in gigawatts as well. In this last year from 2024 to 2025, the total power demand has increased by almost 19.85% but the share of this generation provided by the renewables has only increased by 7.13%. To achieve the target of 95% of power production from low carbon sources, the rate of increase of renewables not only has to catch up to 20% but also exceed it.

The quantitative analysis above indicates that it will be a significant challenge for the government to achieve its 'Clean Power Targets of 2030'.

Key Insights

Apart from the above objectives, there are other significant insights that are gained through this data analysis:

- The most significant increase has been seen in the power generated by wind. From a mere 1.29% in 2012 to 35.36% of total power generation in 2025.
- Power generated by coal has gone down from 21.57% in 2010 to 0.5% in 2024 to 0% in 2025.
- Power generated by biomass has gone up from 0% of the total fuel mix to 7.92% in 2025.

- Power imported into UK has gone up from 3.17% in 2010 to 17.20% in 2025.

Conclusion

There has been significant change in the electricity supply in UK in the past 16 years as can be seen from the above analysis. The power generation in UK has constantly been moving towards more carbon neutral options. Biomass, solar and increasing usage of wind in power generation is moving UK forward in its journey towards not only achieving its low-carbon emission goals but also towards self-sufficiency in power generation. The fuel mix is much more diverse than before and this diversity is expected to rise in the future as well.

Limitations and Some Questions to Ponder On

According to the Clean Power Mission 2030, apart from setting targets for low carbon emissions and increasing the usage of renewable sources in power generation, it also has set targets for further electrification of industry in UK which I have not considered above.

I have looked at two of the targets of the Clean Power Mission 2030 and made my objective to analyse **only the NESO's historic data** to focus on decarbonisation and the increase in renewables. Hence, my horizon has narrowed.

For a more holistic view (which is out of the scope of this project), to consider if UK is on track to achieving the above two targets, the whole of the Clean Power Mission needs to be considered. The suggestions given by NESO given in 2025 need to be taken into account. For a wider perspective, questions need to be asked, some of which are:

- What practical steps were taken in 2025 to achieve Clear Power Mission 2030 targets?
- How effective have these targets been?
- What projects for renewables and net zero generation have been started?
- How have they progressed over the last year?
- What other projects are in the pipeline and what are their respective timelines?

The decarbonisation process of UK's power generation is a significant challenge. There are two reasons for this. Firstly, the existing usage of fossil fuel has to be replaced.

Secondly, low carbon and net zero sources have to further fulfill the increasing internal electricity demand. This demand, which had in the past been decreasing due to efficiency

and conservation of energy, is expected to rise because of the government's drive towards electrification or setting up of more data centers, etc